

Practice Problems for EE 595H

Problem 1

Alex is an avid sports fan who watches football games regularly. His friend has observed that whether Alex watches a football game on a given day depends on the previous two games he watched, with the following probabilities:

- $7/8$ if the last two games he watched were both football games,
 - $1/2$ if exactly one of the last two games was a football game,
 - $1/8$ if neither of the last two games was a football game.
- (a) Alex watched four games last week. If the first two were football, what is the probability that the fourth one was also a football game?
- (b) What is the long-run probability that Alex will watch two football games in a row?

Problem 2

On a given day, a retired person keeps himself busy with only one of the following activities:

- Activity 1: Reading
- Activity 2: Gardening
- Activity 3: Watching TV

Define the random variable

$$X_n = i \quad \text{if the person devotes day } n \text{ to Activity } i.$$

Suppose $\{X_n : n = 1, 2, \dots\}$ forms a Markov chain. The transition probability matrix is

$$P = \begin{bmatrix} 0.3 & 0.25 & 0.45 \\ 0.4 & 0.1 & 0.5 \\ 0.25 & 0.4 & 0.35 \end{bmatrix}.$$

Find the proportion of days the person devotes to each activity.

Table 1: Transition Probabilities and Rewards

State	Action	Next State	Transition Probability	Reward
s_1	a_1	s_1	0.3	2
s_1	a_1	s_2	0.6	4
s_1	a_1	s_3	0.1	1
s_1	a_2	s_2	0.7	5
s_1	a_2	s_3	0.3	3
s_2	a_1	s_1	0.5	1
s_2	a_1	s_3	0.5	2
s_2	a_2	s_3	1.0	7
s_3	a_1	s_1	0.2	-1
s_3	a_1	s_3	0.8	0
s_3	a_2	s_3	1.0	0

Problem 3

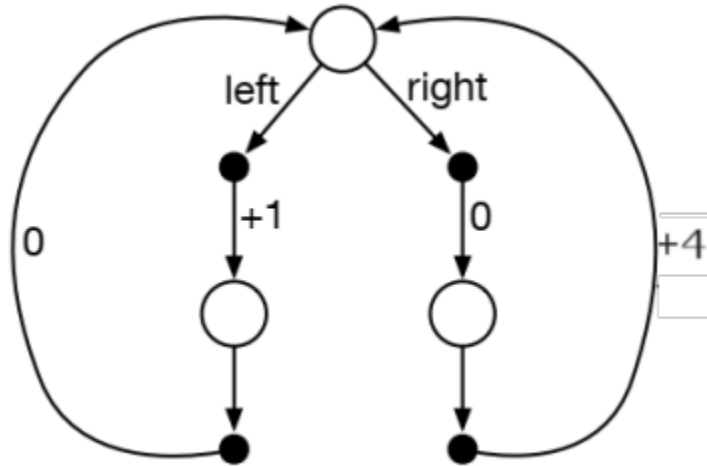
Consider an MDP with three states $S = \{s_1, s_2, s_3\}$ and two actions $A = \{a_1, a_2\}$. The discount factor is $\gamma = 0.9$. The transition probabilities and rewards are given in Table 1.

Initialize $v_0(s) = 0$ for all states. Using Bellman's optimality equation, compute the values $v_1(s)$ and $v_2(s)$. Also find the optimal deterministic policy after two iterations.

Do the computations only up to 2 iterations.

Problem 4

Consider the continuing MDP shown in the following figure. The only decision is made at the top state, where two actions are available: *left* and *right*. The numbers indicate the deterministic rewards received after each action.



There are exactly two deterministic policies, π_{left} and π_{right} .

- What policy is optimal if $\gamma = 0$?
- What policy is optimal if $\gamma = 0.5$?
- What policy is optimal if $\gamma = 0.25$?